

15. (2.25 points)

For each of the scenarios in parts a., b., and c. below, briefly describe:

- i. The likely impact on the loss data,
- ii. The adjustment that should be made by incorporating expert opinion, and
- iii. The impact on IBNR estimates if no adjustments are made.

a. (0.75 point)

Claims staffing level has increased over the last year relative to the previous four years.

b. (0.75 point)

The company's property portfolio has grown rapidly in catastrophe-prone areas.

c. (0.75 point)

A large claim reserved at \$10M ultimately settles favorably for \$1M.

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EXAM 7 SPRING 2018 SAMPLE ANSWERS AND EXAMINER'S REPORT

SPRING 2018 EXAM 7, QUESTION 15	
TOTAL POINT VALUE: 2.25	LEARNING OBJECTIVE(S): A8
SAMPLE ANSWERS	
Part a: 0.75 point	
<u>Sample 1</u>	
i. The claim reporting pattern has changed. With more claims staff, adjusters are able to spend more time setting initial case reserves than in prior years. The latest year case reserves are probably more sufficient so we should expect less development in the future. ii. We should incorporate expert opinion by selecting a LDF for the latest year that reflects the expected change in loss reporting (a lower LDF). iii. If no adjustments are made then IBNR will be too high.	
<u>Sample 2</u>	
i. Speedup in claim payment for these years. ii. Reduce the expected LDF. iii. IBNR would be overstated.	
Part b: 0.75 point	
<u>Sample 1</u>	
i. Rapid growth in CAT prone areas may or may not appear in the loss data. If no catastrophes have occurred, then the loss data will be unaffected but the risk will still exist. ii. We should incorporate expert opinion by using a CAT model to have a better understanding of the loss experience and development should be analyzed separately from non-CAT losses. The CAT model results can be used to set reserves for events that occurred but we are unsure still of the total loss. iii. If no adjustments are made, IBNR would be inappropriate as non-CAT and CAT losses can have very different reporting patterns, or risk from CATs won't be reflected in IBNR.	
<u>Sample 2</u>	
i. Growth in premium, volatility in losses in recent years. ii. Calculate IBNR for property cat losses separately. iii. Year to year volatility in IBNR.	
Part c: 0.75 point	
<u>Sample 1</u>	
i. The loss data will be impacted by showing negative development from one large claim. This could be leveraging a development pattern. ii. This claim should be removed from the data since it is an anomaly. We don't want it to affect the age-to-age factors or IBNR estimates. iii. If no adjustments are made and the claim is left in the data, the selected age-to-age factor will be lower than it should be and IBNR will be too low.	

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Sample 2

- i. A negative development b/w the two ages that the settlement happens.
- ii. Remove this outlier from the loss data.
- iii. IBNR would be understated.

EXAMINER'S REPORT

Candidates were expected to be able to identify the impact of the situations on the loss data and on the IBNR. They were also expected to propose an adjustment using expert opinion that corrected the impact on IBNR.

Part a

Candidates were expected to know that more claim staff means that reserves will show a shorter development pattern, and also the impact of a shorter development pattern on the IBNR calculated with the chain-ladder method.

Common mistakes included:

- Identifying claim adjusters as the expert opinion, since they are not considered experts on the same level as reserving actuaries with respect to IBNR and would not be making adjustments to the data.

Part b

Candidates were expected to acknowledge that including catastrophe losses with attritional losses for reserving purposes would create distortions in the resulting IBNR.

Common mistakes included:

- Stating that catastrophe losses should be removed from the triangle, but without mentioning that they should be either added back through a cat loading or analyzed separately with a cat model.

Part c

Candidates were expected to identify that the downward development would have an impact on the age-to-age factor at the time of settlement, which in turn would create understated IBNR if the downward development is identified as an outlier.

Common mistakes included:

- Stating that no adjustment is necessary because the claim is closed, thus will not further develop and that this is included in the calculated IBNR. This ignores the impact on the LDF and the expected development of other claims.
- Proposing as an adjustment to lower the case reserve for that claim, assuming that the downward development was not already reflected in the triangle. This is not a valid assumption.